

MEDIATEK

MT8676 Yocto eCall

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Version History

Version	Date	Author	Description
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1 eCall

1.1 Overview

1.1.1 Introduction

eCall is an in-vehicle road safety system which automatically calls the emergency services in case of a serious accident. As soon as the eCall sensors register a severe impact on a vehicle or a call is initiated manually, eCall In-Vehicle System (IVS) establishes e112 voice connection with the relevant Public Safety Answering Point (PSAP).

Send a Minimum Set of Data (MSD) over the voice connection to PSAP, that includes accurate geo-location data

In April 2015, the European Parliament made it mandatory for all new models of cars to be equipped with eCall technology from 31 March 2018 onward. The eCall system overview is depicted in Figure 1-1.

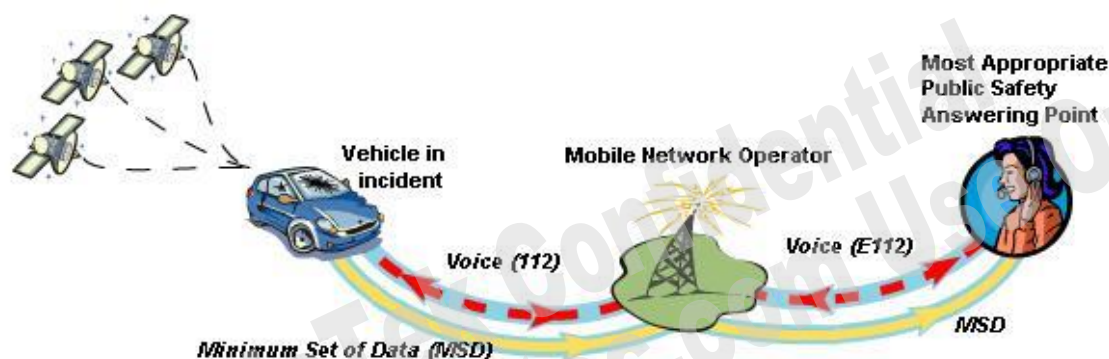


Figure 1-1. eCall system overview

1.1.2 Specification Introduction

Basically there are two different types of eCall specification, one is for modem protocol such as 3GPP specification, the other is for high-layer application specification such EN specification. Generally MediaTek will cover modem protocol part and provide SDK for customer to integrate high-layer application specification with their own application.

The tables below simply show the different specifications. For detailed information, please refer to the original specifications from official websites.

Table 1-1. CS eCall functional specification

Spec Name	Description
ETSI TS 126.267 (3GPP TS 26.267)	General description
ETSI TS 126.268 (3GPP TS 26.268)	ANSI-C reference code
ETSI TS 127.007 (3GPP TS 27.007)	6.27 Initiate eCall +CECALL
ETSI TS 122.101 (3GPP TS 22.101)	10.7 Transfer of data during emergency calls
ETSI TS 124.008 (3GPP TS 24.008)	4.4.7 eCall inactivity procedure
3GPP TS31.102	SIM related requirement for eCall
ETSI TS 102 936-1	5. UMTS network access eCall NAD conformance tests 6.GSM network access eCall NAD conformance tests

Spec Name	Description
EN 16062	eCall high level application requirements (HLAP)
EN 16072	PE eCall Operating Requirements
EN 15722	eCall minimum set of data (MSD)
GOST 33465-2015	Protocols of data exchange between in-vehicle emergency call device/system and emergency response system infrastructure
GOST 33464-2015	In-vehicle emergency call device/system. General technical requirement

Table 1-2. CS eCall test specification

Spec Name	Description
ETSI TS 126.269 (3GPP TS 26.269)	eCall Data Transfer; In-band modem solution; conformance test
ETSI TS134 123-1	13.3 eCall Emergency Call Procedures
ETSI TS 151 010-1	Conformance specification 26.9.6a Structured Calls/eCall
ETSI TS103 412	Pan-European eCall end to end and in-band modem conformance testing; Prose test specification
EN16454	Intelligent transport systems –ESafety- Ecall end to end conformance testing
GOST 33467-2015	Functional test methods of in-vehicle emergency call device/system and data transfer protocols
GOST 33470-2015	Test methods for wireless communication modules of in-vehicle emergency call system
ETSI TS 103 428	eCall HLAP Interoperability Testing

Table 1-3. IMS eCall functional specification

Spec Name	Description
TS24.229	eCall over IMS(NG-eCall) general spec
RFC8147	NG-eCall SIP part spec
TS23.122	NG-eCall eCall only mode
TS24.301	EMM requirement for eCall only mode
TS23.401	eCall only mode
TS24.008	MM requirement for eCall only mode
TS23.216	NG-eCall SRVCC
TS23.167	eCall over IMS(NG-eCall) general spec
TS27.007	AT CMD of eCall
TS31.102	SIM-related requirement for eCall

Table 1-4. IMS eCall test specification

Spec Name	Description
TS34.229-1	SIP layer test spec of NG-eCall
TS36.523	UE conformance test spec of NG-eCall

1.2 Architecture/Process Overview

1.2.1 eCall Architecture

The Software architecture is depicted in Figure 1-2.

Currently, the MTK solution supports eCall setup and in-band modem TX/RX, compliant with ETSI/3GPP standards (green rectangle), and provides proprietary ML interface for invoking eCall RIL commands. Customers need to integrate the eCall platform and application (red rectangle) with the eCall ML interface and use the product to pass eCall test cases.

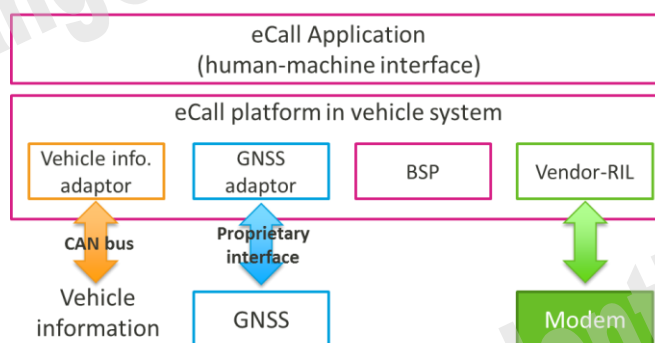


Figure 1-2. eCall SW architecture

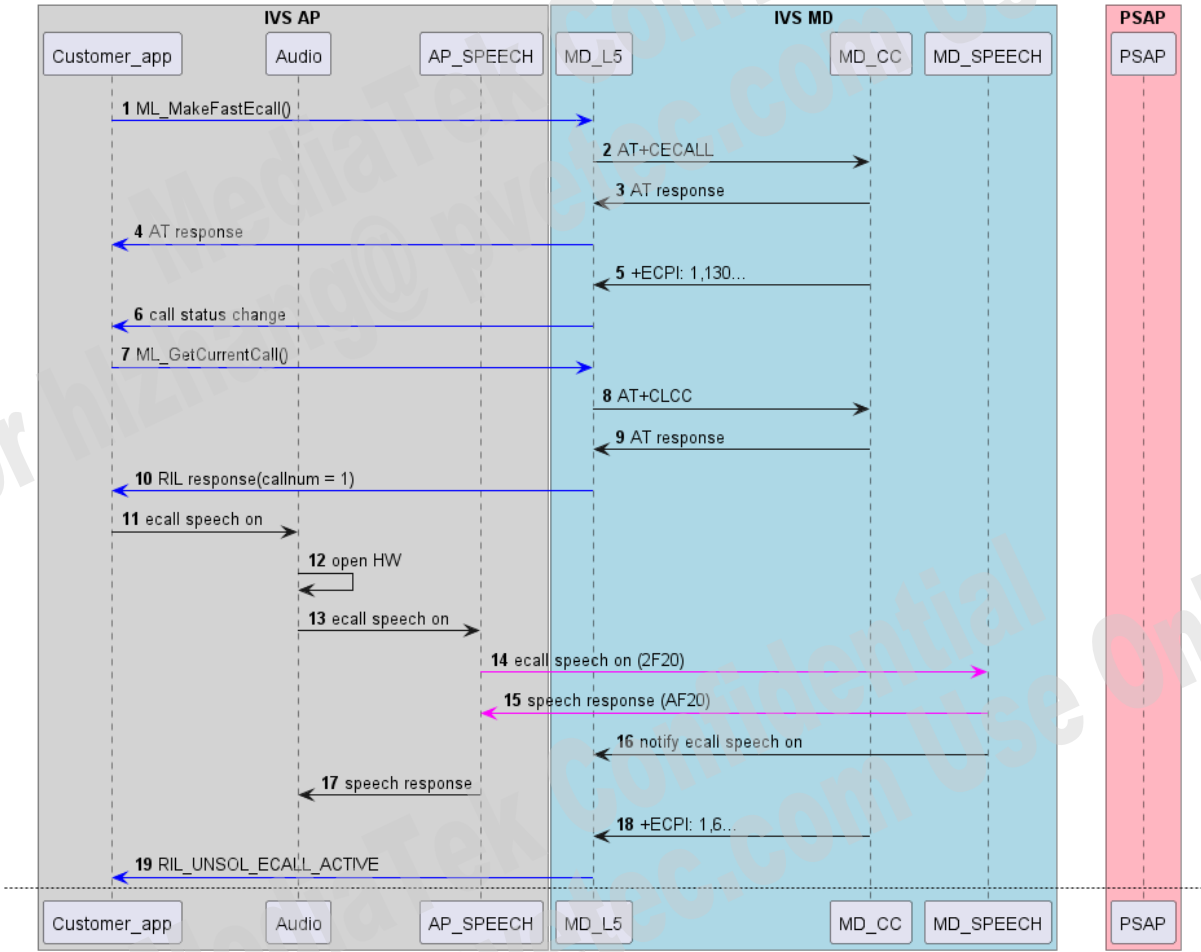
1.2.2 eCall Flow

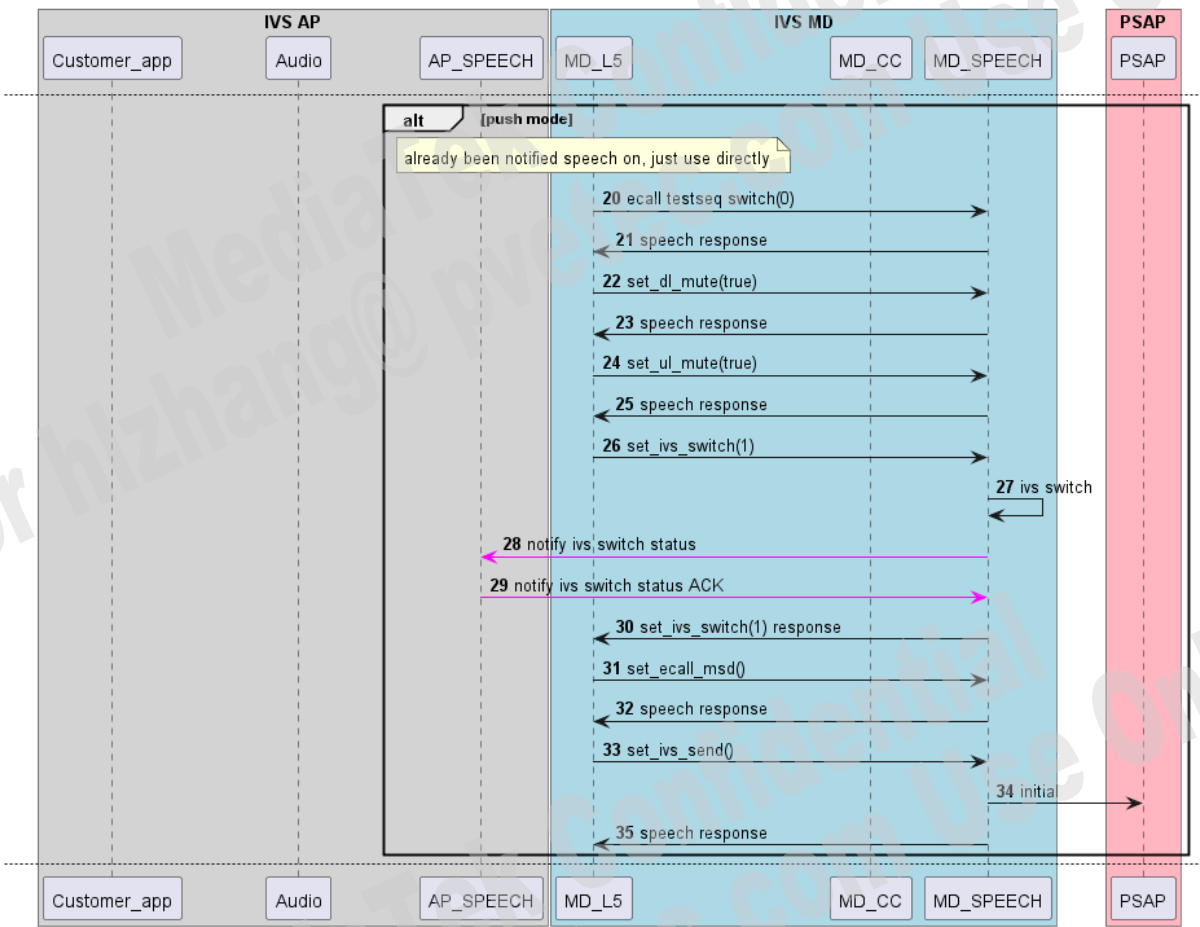
Although there are different eCall scenarios, the keys steps are the same. Basically, customer APP needs to handle eCall flow according to the following steps:

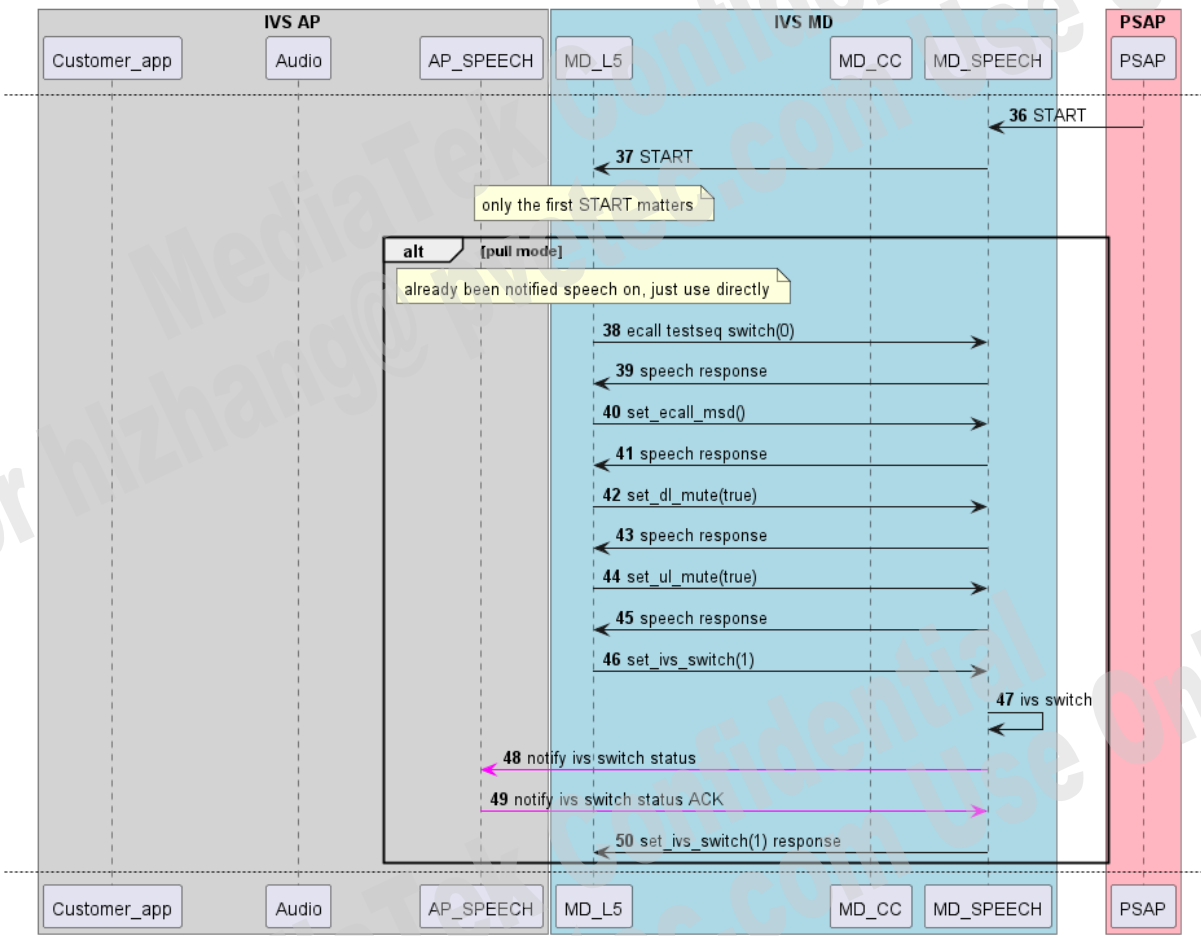
1. Initiate an ML_MakeFastEcall
2. Maintain eCall timer of IVS side which is specified in EN16062
3. Issue RIL_REQUEST_ECALL_RESET_IVS after receiving RIL_UNSOL_ECALL_ALACK_POSITIVE_RECEIVED. Then IVS and PSAP can resume voice call connection and talk to each other
4. Upon receiving RIL_UNSOL_ECALL_DISCONNECTED or RIL_UNSOL_ECALL_ABNORMAL_HANGUP, disconnect the call. The PSAP uses this method to instruct the IVS to disconnect the call..

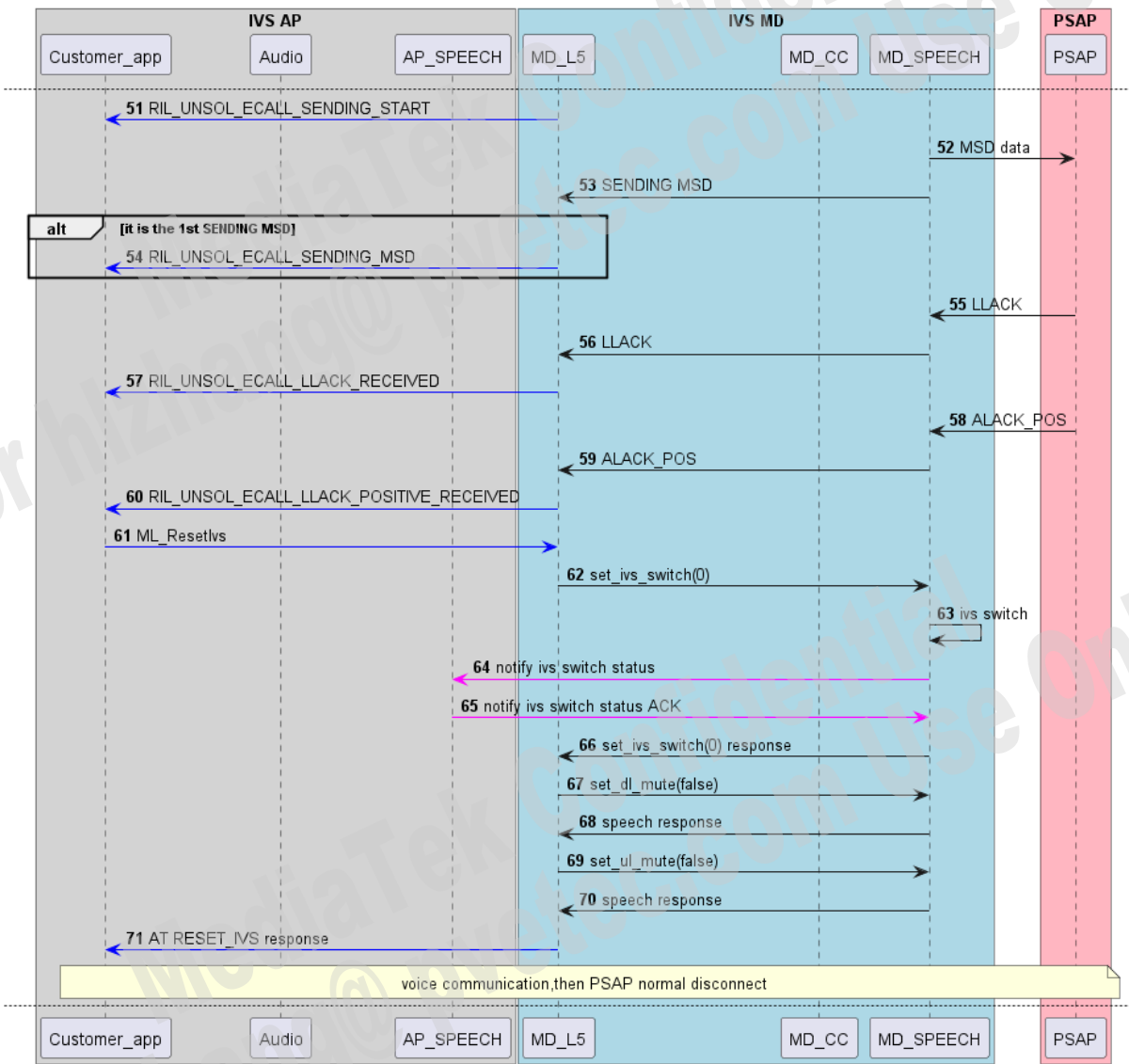
Figure 1-3 shows the detailed description of eCall flow sequence.

mt8676 eCall overall normal sequence









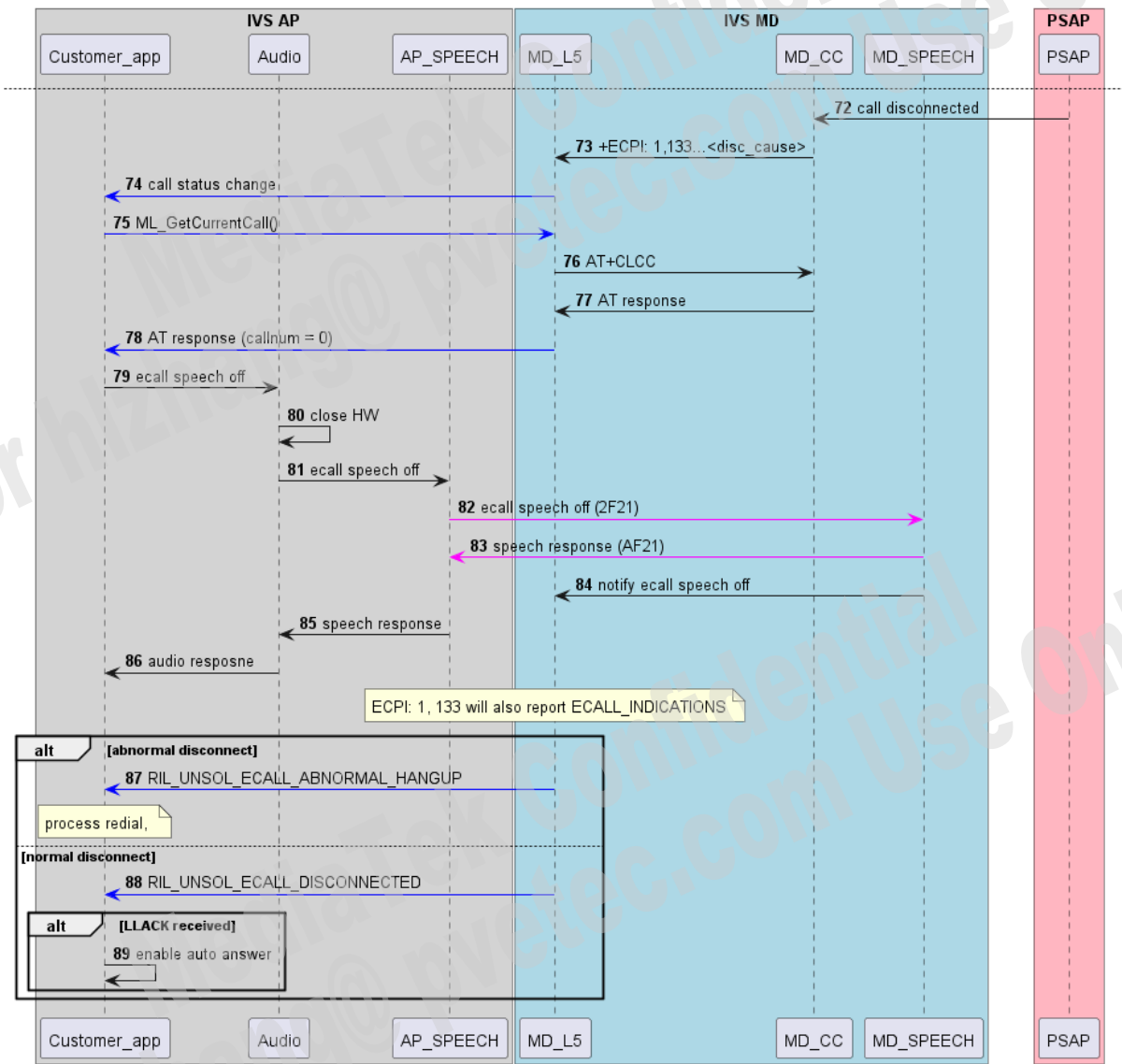


Figure 1-3. eCall flow chart of PSAP

1.2.3 eCall API Usage

The eCall APIs shown in Table 1-5 are planned to develop on MT8676. If there are updates in the subsequent development, updated documents will be provided..

Table 1-5. eCall control interface description

Interface/Struct	Description
<pre>typedef struct { int32_t call_id; uint32_t length; unsigned char msd_data[ML_EC_CALL_MSD_MAX_LENGTH]; }ml_ecall_set_msd;</pre>	Struct of MSD information

Interface/Struct	Description
<pre>typedef struct { int32_t arg_num; int32_t type; char address[128]; }ml_ecall_set_num;</pre>	Struct of test number/reconfiguration number information
<pre>typedef enum { ML_EMER_CAT_MANUAL_ECALL = 1, ML_EMER_CAT_AUTO_ECALL = 2, }ml_ecall_category;</pre>	Enum of eCall category
<pre>typedef enum { ML_ECALL_TEST = 1, ML_ECALL_EMERGENCY = 2, ML_ECALL_RECONFIG = 3, }ml_ecall_variant;</pre>	Enum of eCall variant
<pre>typedef enum{ ML_DOMAIN_AUTO = 0, /* Automatic mode - LTE(IMS), WG(CS), 1x(C2K) */ ML_DOMAIN_CS_ONLY = 1, /* CS domain only - WG(CS) */ ML_DOMAIN_3GPP_ONLY = 2, /* 3GPP only - LTE(IMS), WG(CS) */ ML_DOMAIN_3GPP2 = 3, /* 3GPP2 only - 1x(C2K) */ ML_DOMAIN_IMS_1xCS = 4, /* IMS and 1x CS only - LTE(IMS), 1x(C2K) */ ML_DOMAIN_CS_1x = 5, /* WG CS and 1x CS only - WG(CS), 1x(C2K) */ ML_DOMAIN_IMS_ONLY = 6, /* only IMS call allowed */ }ml_ecall_domain;</pre>	Enum of radio domain when make eCall
<pre>typedef struct{ ml_ecall_category ecall_cat; ml_ecall_variant ecall_variant; char address[20]; uint32_t length; unsigned char msd_data[ML_ECALL_MSD_MAX_LENGTH]; ml_ecall_domain domain; }ml_ecall_req_msg;</pre>	Struct of input parameter when requesting to make an eCall.

Interface/Struct	Description
<pre>typedef struct{ int32_t data1; int32_t data2; int32_t data3; int32_t data4; }ml_ecall_pri;</pre>	eCall priority parameter structure. data1>data2>data3>data4. The input parameters should be 1, 2, 3, 4, which means: 1: eCall URI set by the customer 2: eCall URI saved by USIM 3: eCall number set by the customer 4: eCall number saved by USIM
<pre>typedef enum{ E_ML_ECALL_SENDING_START = 1, E_ML_ECALL_SENDING_MSD = 2, E_ML_ECALL_LACK_RECEIVED = 3, E_ML_ECALL_ALACK_POSITIVE_RECEIVED = 4, E_ML_ECALL_ALACK_CLEARDOWN_RECEIVED = 5, E_ML_ECALL_DIALING = 9, E_ML_ECALL_ALERTING = 10, E_ML_ECALL_ACTIVE = 11, E_ML_ECALL_DISCONNECTED = 12, E_ML_ECALL_IMS_ACTIVE = 13, E_ML_ECALL_IMS_DISCONNECTED = 14, E_ML_ECALL_ABNORMAL_HANGUP=15, E_ML_ECALL_IMS_MSD_ACK = 20, E_ML_ECALL_IMS_UPDATE_MSD = 21, E_ML_ECALL_IMS_IN_BAND_TRANSFER = 22, E_ML_ECALL_IMS_MSD_NACK = 23, E_ML_ECALL_IMS_SRVCC = 24, E_ML_ECALL_ONLY_DEREGISTRATION = 31, E_ML_ECALL_MAY_DEREGISTER = 32, E_ML_ECALL_PSAP_CALLBACK_START = 40, E_ML_ECALL_PSAP_CALLBACK_IMS_UPDATE_MSD = 41, E_ML_ECALL_T2_TIMEOUT = 52, E_ML_ECALL_T5_TIMEOUT = 55, E_ML_ECALL_T6_TIMEOUT = 56, E_ML_ECALL_T7_TIMEOUT = 57, E_ML_ECALL_UNSPECIFIED = 0xffff, }ML_ECall_Indication;</pre>	Enum of eCall indication type
<pre>typedef struct{ ML_ECall_Indication ind; int call_id;</pre>	Struct of eCall indication

Interface/Struct	Description
} ML_ECALL_IND_T;	
typedef void (*ML_ECALL_MSGCB_T)(ML_ECALL_IND_T *pvsMsg);	Callback function for reporting eCall indication
int32_t ML_EcallIndicationInit(ML_ECALL_MSGCB_T cb_func);	Register the eCall indication callback function, and call back the cb_func function when the eCall status changes. Parameters: In: ML_ECALL_MSGCB_T cb_func
int32_t ML_ResetIvs(void);	Reset eCall state in modem and reconnect audio channel.
int32_t ML_SetMSD(ml_ecall_set_msd* msd);	Set MSD. Parameters: In: ml_ecall_set_msd* msd
int32_t ML_SetTestNumber(ml_ecall_set_num* test_num);	Set test number or URI. Parameters: In: ml_ecall_set_num* test_num
int32_t ML_SetReconfNumber(ml_ecall_set_num* reconf_num);	Set reconfig number or URI. Parameters: In: ml_ecall_set_num* test_num
int32_t ML_MakeFastEcall(ml_ecall_req_msg* msg);	Trigger an eCall and transmit MSD data in one command. Parameters: In: ml_ecall_req_msg* msg
int32_t ML_SetEmsdpri(ml_ecall_pri* pri);	Set priority of test/reconfiguration eCall number/URI The default priority is “1>3>2>4” . Parameters: In: ml_ecall_pri* pri

1.3 eCall Timer

1.3.1 EU CS eCall Timer

In an eCall transaction, there are 10 timers defined in EN 16062. MediaTek has implemented T2, T3, T5, T6, T7, T10 in modem side. T1, T9 should be handled by IVS customer APP; T4 and T8 are PSAP timers and should be handled by PSAP application.

Table 1-6. eCall timers in EN16062

Name Origin	Description Requirements	Value
T1 IVS	Manually initiated eCall(MleC) false triggering cancellation period START: T1 starts as soon as the eCall is manually activated	

Name Origin	Description Requirements	Value
	- STOP: T1 stops when Vehicle occupants cancel the manually triggered eCall transaction. - EXPIRY: Upon expiry of T1 the IVS-NAD shall start call setup	
T2 IVS	IVS Call Cleardown Fallback Timer (CCFT) - START: T2 starts as soon as the IVS-NAD starts with call setup - STOP: T2 stops when the IVS-NAD receives a call clear-down indication from the mobile network or a call clear-down message from the PSAP. - EXPIRY: Upon expiry of T2 the IVS-NAD shall clear down the call	3600 s (1h)
T3 IVS	IVS INITIATION signal duration - START: T3 is started as soon as the IVS-NAD starts sending the INITIATION signal - STOP: T3 stops when the IVS-NAD receives a SEND MSD signal from the PSAP, at which time the IVS-NAD shall stop sending the INITIATION signal. - EXPIRY: Upon expiry of T3 the IVS-NAD shall stop sending the INITIATION signal	2 s
T4 PSAP	PSAP wait for INITIATION signal period - START: T4 starts as soon as the PSAP eCall modem has answered the call - STOP: T4 stops when the PSAP eCall modem detects an INITIATION signal send by the IVS. - EXPIRY: Upon expiry of T4, the PSAP eCall modem shall route the call to a PSAP operator	5 s
T5 IVS	IVS wait for SEND MSD period - START: T5 starts as soon as the IVS-NAD received notification that the call is first answered - STOP: T5 stops when the IVS-NAD detects a SEND MSD signal sent by the PSAP. - EXPIRY: Upon expiry of T5 the IVS-NAD shall reconnect the IVS audio system and terminate eCall specific behavior(i.e. it shall not proceed with the sending of MSD data) until requested to do otherwise.	5 s
T6 IVS	IVS wait for AL-ACK period - START: T6 starts as soon as the IVS-NAD has received LL-ACK - STOP: T6 stops when the IVS-NAD receives an AL-ACK message - EXPIRY: Upon expiry of T6, the IVS-NAD shall mark the transfer of the MSD as unsuccessful and reconnect the IVS audio system and terminate eCall specific behavior until requested to do otherwise	5 s
T7 IVS	IVS MSD maximum transmission time - START: T7 starts as soon as the IVS-NAD starts sending the MSD data - STOP: T7 stops when the IVS-NAD receives an LL-ACK message	20 s

Name Origin	Description Requirements	Value
	EXPIRY: Upon expiry of T7, the IVS-NAD shall mark the transfer of the MSD as unsuccessful and reconnect the IVS audio system and terminate eCall specific behavior until requested to do otherwise	
T8 PSAP	PSAP MSD maximum reception time - START: T8 starts as soon as the PSAP starts sending the SEND MSD signal - STOP: T8 stops when the PSAP eCall modem receives a valid MSD (reception being acknowledged by sending an LL-ACK) - EXPIRY: Upon expiry of T8, the PSAP eCall modem shall route the call to a PSAP operator	20 s
T9 IVS	IVS NAD minimum network registration period - START: T9 starts as soon as the IVS-NAD clears down a call, or gets notified that a call has been cleared down in accordance with EN 16072 Clause 7.17.3 - STOP: T9 is uninterruptable; until T9 expires the IVS-NAD shall remain registered on the serving network, and remain available to receive calls from the PSAP and rescue workers - EXPIRY: Upon expiry of T9, the IVS-NAD may deregister from the serving network (see T10)	3600 s
T10 IVS	IVS NAD network 'Deregistration Fallback Timer' (DFT) - START: T10 starts as soon as the IVS-NAD clears down a call, or gets notified that a call has been cleared down - STOP: T10 stops if the IVS-NAD receives or makes a new call - EXPIRY: Upon expiry of T10, the IVS-NAD shall deregister itself from the serving network	12 h

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